

PERSONAL DATA SCIENCE PROJECT · 2026 REVISION

Ten Years of Chicago's Carbon Ledger

Revisiting a 2023 team capstone with fresh data, a rigorously tested question, and Claude Code as a working collaborator.



From Team Capstone to Personal Case Study

2023 — ORIGINAL

- 7 collaborators, one semester
- 2014-2020 snapshot, 17,728 rows
- A t-test that was defined but never run
- A statistically invalid zip-code correlation
- Regression, a classifier, and a map with no shared throughline

2026 — THIS REVISION

- Solo revision, built with Claude Code
- 2014-2023 refresh, 28,329 rows
- Every bug found by actually running the notebook
- Two panel-based questions, tested properly
- Every chart and claim verified before shipping

Why Revisit This Now?

In September 2026, the EPA repealed its rule limiting greenhouse-gas emissions from power plants — rolling back federal climate policy by roughly a decade.

City ordinances like Chicago's building benchmarking law are increasingly one of the few levers left. This analysis asks what the emissions-intensity trend actually looks like among the buildings it covers - not whether the ordinance caused it.

A Decade of Data, Not Six Years

Original snapshot

17,728

rows · 2014-2020



Refreshed Sept 2026

28,329

rows · 2014-2023

Same source, three more years of city reporting, +10,601 rows.



What Actually Running the Code Found

Bug	Why it mattered
Numbers like "104,849" loaded as text	Every downstream calculation on floor area, electricity use, and GHG emissions would have silently run on strings, not numbers
geopandas import commented out	The map cells later in the notebook would crash with NameError on a fresh run
Row-by-row loops with chained indexing	Slow, warning-prone cleaning of neighborhoods, spelling fixes, and regions
Zip code correlation and an unrun t-test	Both original "Hypothesis Testing" results were statistically invalid or simply incomplete



Two Questions, Tested Properly

1

Has GHG intensity among buildings in the benchmarking dataset actually declined — or does that just reflect which buildings happen to report each year?

2

Among buildings tracked for years, are they individually cutting their own emissions intensity?



Finding 1: The Decline Holds Up

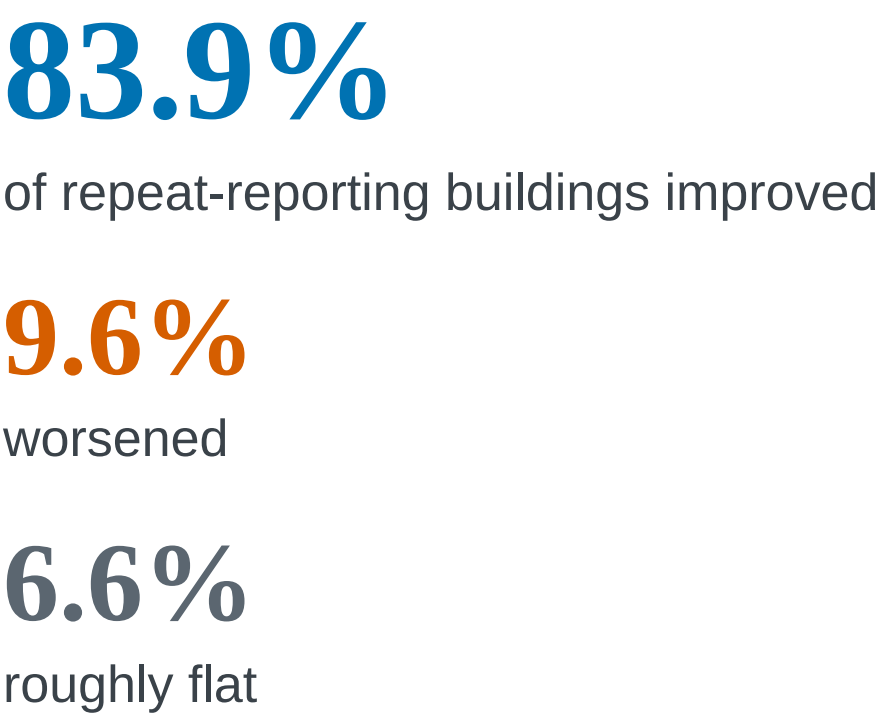
 Line chart comparing median GHG intensity 2014-2023 for all reporting buildings versus a fixed panel of repeat reporters; the two lines nearly overlap, both declining from about 13 to 6.2 kg CO2e per square foot

Median GHG intensity, all reporters vs. a fixed panel of repeat reporters (4+ years of data) — nearly identical.



Finding 2: Most Buildings Are Improving on Their Own

 Histogram of each building's per-year GHG intensity trend, colored blue for improving, gray for flat, and orange for worsening; most bars are blue





Same Story, Mapped

 Animated map of Chicago with one frame per year from 2014 to 2023, showing every reporting building colored by that year's GHG intensity on a light-to-dark blue scale; the point cloud fills in and lightens over time

Two signs of the same decline, visible at once: the point cloud fills in, and the dots trend lighter.

240 → 2,572

buildings reporting, 2014 vs. 2023

Reuses `chicago_outline.shp`, a shapefile that sat unused in the repo after the original map section was cut.

The Trend Is Real — But It's Not That Simple

–36%

2014 → 2016

May be inflated by 2014's record-cold winter

–3.6%

2016 → 2019

Three years of essentially flat progress

–16%

2019 → 2020

A one-year cliff — pandemic vacancy, not retrofits

–7.5%

2020 → 2023

Slow, real, but modest since

Federal power-plant rules were repealed in 2026, partly on the premise that cities can carry more of the climate burden. Buildings in this dataset show a real decline in GHG intensity - but this analysis has no comparison group, so it can't isolate the ordinance's own effect from other trends, and the earliest, largest drop coincides with an unusually cold 2014 winter.

The trend is real. Whether the ordinance caused it is still an open question.

How Claude Code Helped — and What I Directed

I SET THE DIRECTION

- Chose which research questions were worth asking
- Decided what to prune versus keep, for one clear throughline
- Caught that the panel barely differed from the raw average — and that mattered
- Asked for a map that showed the trend, not just decoration

CLAUDE CODE EXECUTED & VERIFIED

- Found the thousands-separator and missing-import bugs by running the code, not just reading it
- Re-ran the full notebook end to end after every single change
- Applied a colorblind-safe, single-axis chart design system
- Was transparent about a network block instead of fabricating data



A Real Exchange From This Project

Me:

"What if we make a map visual that does show the trend across time — like a video?"

Claude Code:

Built an animated map reusing the shapefile that had sat unused since the original project — one frame per year, colored by GHG intensity, verified by executing the notebook and reviewing every frame before shipping it.



Skills This Project Demonstrates

- Data Cleaning & Validation

- Reproducible Analysis

- Panel & Composition-Bias Reasoning

- AI-Assisted Engineering

- Data Visualization Design

- Git & Version Control

Explore the Full Analysis

github.com/zeal-harshi-vetrivel/idscprojectv2

Data cleaning, a panel-based trend analysis, an animated map, and a full commit history showing how this project — and the way I use AI tools — has grown.